

Practice Quiz: Body Science — Module 02: Agents

Name: ____ Date: ____

Part A: Multiple Choice

1. In the context of your body, which of the following can act like an "agent"?

A) Only your whole brain B) Individual cells — like white blood cells that detect threats, move toward them, and attack C) Only you as a whole person D) Only robots and computers

1. A white blood cell detects a bacteria, chases it through your bloodstream, and destroys it. This white blood cell is acting like an agent because:

A) Someone told it what to do B) It senses its environment, has a goal (destroy the invader), and takes action to achieve it C) It is just floating randomly D) It has a brain

1. Your heart beats faster when you're scared even before you consciously decide to be afraid. This shows that:

A) Your heart has no connection to your brain B) Some body agents act on predictions automatically, without needing your conscious awareness C) Fear is not real D) Your heart is broken

1. What is the difference between a cell acting as an agent and you acting as an agent?

A) There is no difference at all B) A cell responds to chemical and physical signals at a tiny scale; you respond to sights, sounds, and social situations at a larger scale, but both sense, predict, and act C) Cells are not agents D) Only humans can be agents

1. Your gut has its own network of neurons, sometimes called the "second brain." This shows that:

A) You have two brains that fight each other B) Agency is distributed — different parts of your body make their own predictions and decisions C) The gut does not interact with the rest of the body D) Neurons only exist in your head

1. Bacteria in your gut help you digest food. In return, you provide them with a warm, food-rich environment. This relationship is an example of:

A) Two agents with conflicting goals B) Two agents (you and the bacteria) whose goals align — both benefit C) A system with no agents D) Something that has nothing to do with Active Inference

1. When you stub your toe, pain signals travel to your brain, and you quickly pull your foot away. The "agent" that pulled your foot away was:

A) The toe itself B) A reflex arc in your spinal cord — an agent that acted before your brain even fully processed the pain C) Someone else moving your foot D) A random muscle twitch

Part B: Short Answer

1. Explain how a **white blood cell** acts like an agent. What does it sense? What is its goal? What actions does it take? What happens when it encounters something it hasn't seen before (a prediction error)?

2. Your body contains trillions of cells, many of which act like tiny agents. But you also act as one unified agent when you make a decision. How do all these **tiny agents work together** to create a single "you"? Give an example.

3. Compare how **you** respond to a cold day (putting on a jacket) with how your **body cells** respond (shivering, constricting blood vessels). Both are agents taking action. What are the similarities and differences in how these agents sense, predict, and act?